

Toda extensión sobre un entero es un módulo finitamente generado

Teorema 1. *Sea $A \subset B$ una extensión de anillos y $b \in B$ un elemento entero sobre A*

$\implies A[b]$ es un A -módulo finitamente generado.

Demostración: Como b es entero sobre A , existe un polinomio mónico $p \in A[x]$ dado por $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ tal que $p(b) = 0$. Por tanto,

$$b^n = -a_{n-1}b^{n-1} - \dots - a_0 \implies \forall m \geq n : b^m b^{m-n} = -a_{n-1}b^{n-1}b^{m-n} - \dots - a_0b^{m-n}.$$

Entonces, $1, b, b^2, \dots, b^{n-1}$ generan $A[b]$ como A -módulo. ■

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